CASE STUDY

# Chorismate mutase (3): umbrella sampling

> Updated for EasyHybrid 3.2.0 (2026-10-26). The figures and numbers come from runs with the 3.2.0 development version.

The scan and the NEB (part 1, part 2) give potential energies of single structures. Umbrella sampling (US) gives the **free energy** along the reaction coordinate, the potential of mean force (PMF), averaged over the motion of the enzyme at 300 K. This page runs it twice: a first protocol that the Umbrella Sampling Browser shows to be inadequate at the barrier, and a corrected one.

Each window is a QC/MM molecular dynamics run of the 2388 free atoms (PM6/AMBER, about 0.03 s per step on one core): 10 ps of equilibration and 20 ps of data collection per window.

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## 1. First protocol: the windows of the 0.1 Å scan, k = 600

**Simulate → Umbrella Sampling**:

| Field | Value |
|---|---|
| Input Type | From Trajectory (parallel): `CM_scan.ptGeo` of part 1 (32 structures, RC from −1.68 to +1.43 Å) |
| Reaction Coordinate 1 | read from the scan: multiple distance *4 atoms C5–O7 / C9–C1 |
| Force Constant | 600 kJ mol$^{-1}$ Å$^{-2}$ |
| Number of CPUs | 10 |
| Integrator / Temperature / Time Step | Langevin / 300 K / 0.001 ps |
| Equilibration / Data Collection | 10000 / 20000 steps |

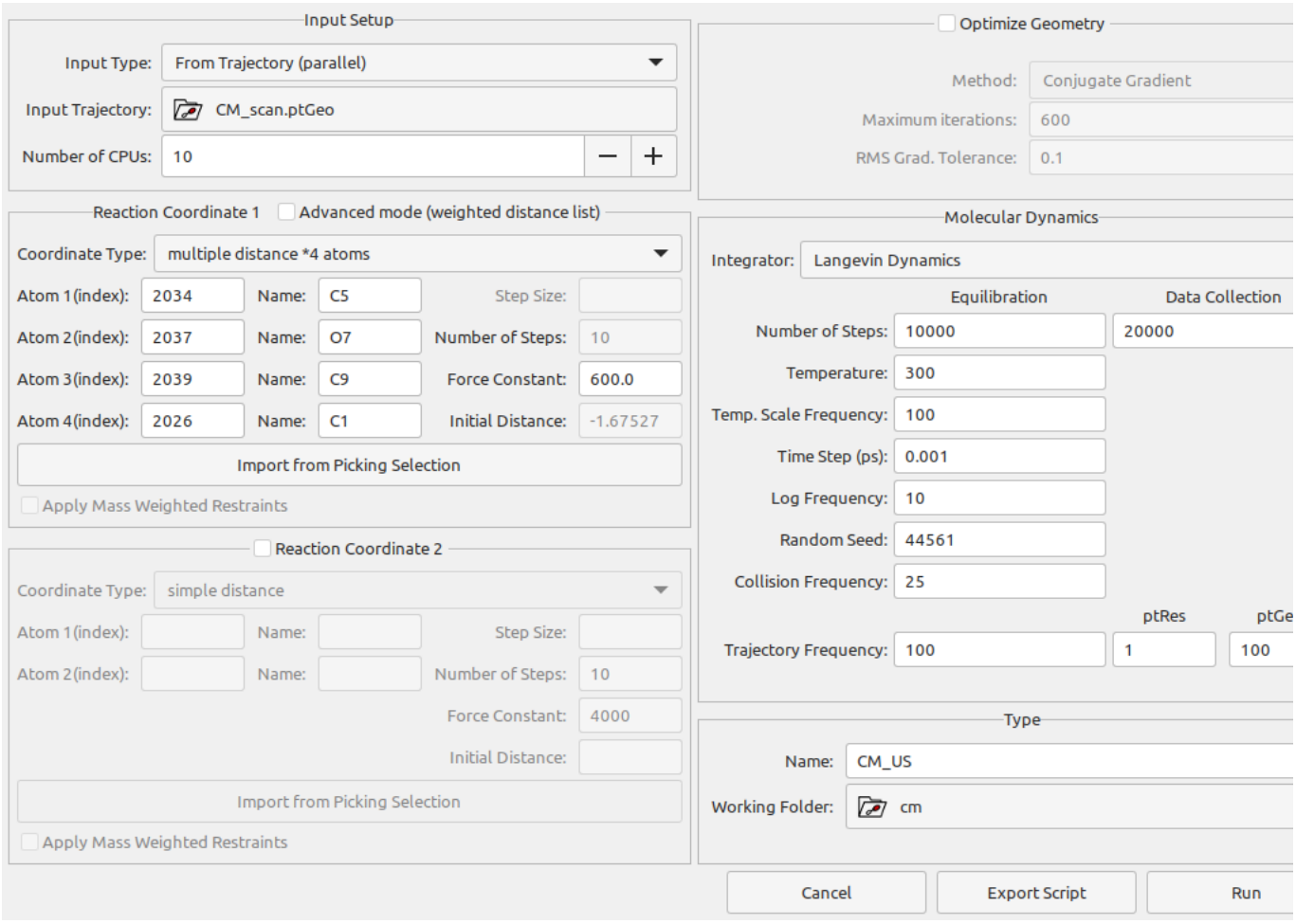



*Figure 1. Windows from the 32 structures of the scan.*

On a flat surface k = 600 spreads each window by σ = 0.046 Å, enough for windows 0.1 Å apart to overlap. The run took 1.5 h on 10 cores.

**Check it in the Umbrella Sampling Browser** (**Analysis → Umbrella Sampling Browser...**):

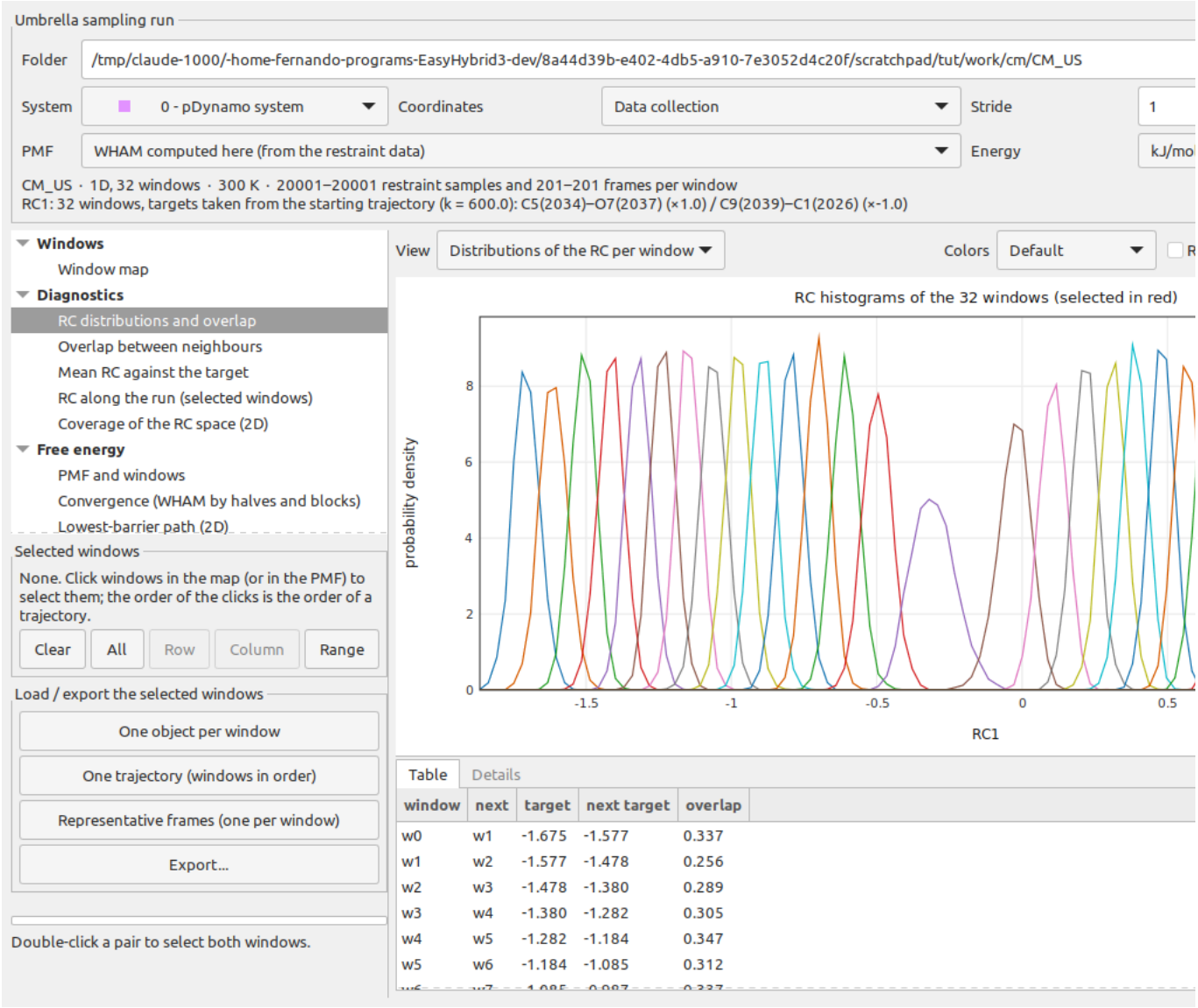



*Figure 2. RC distributions of the 32 windows: a gap around RC = −0.3 to 0.*

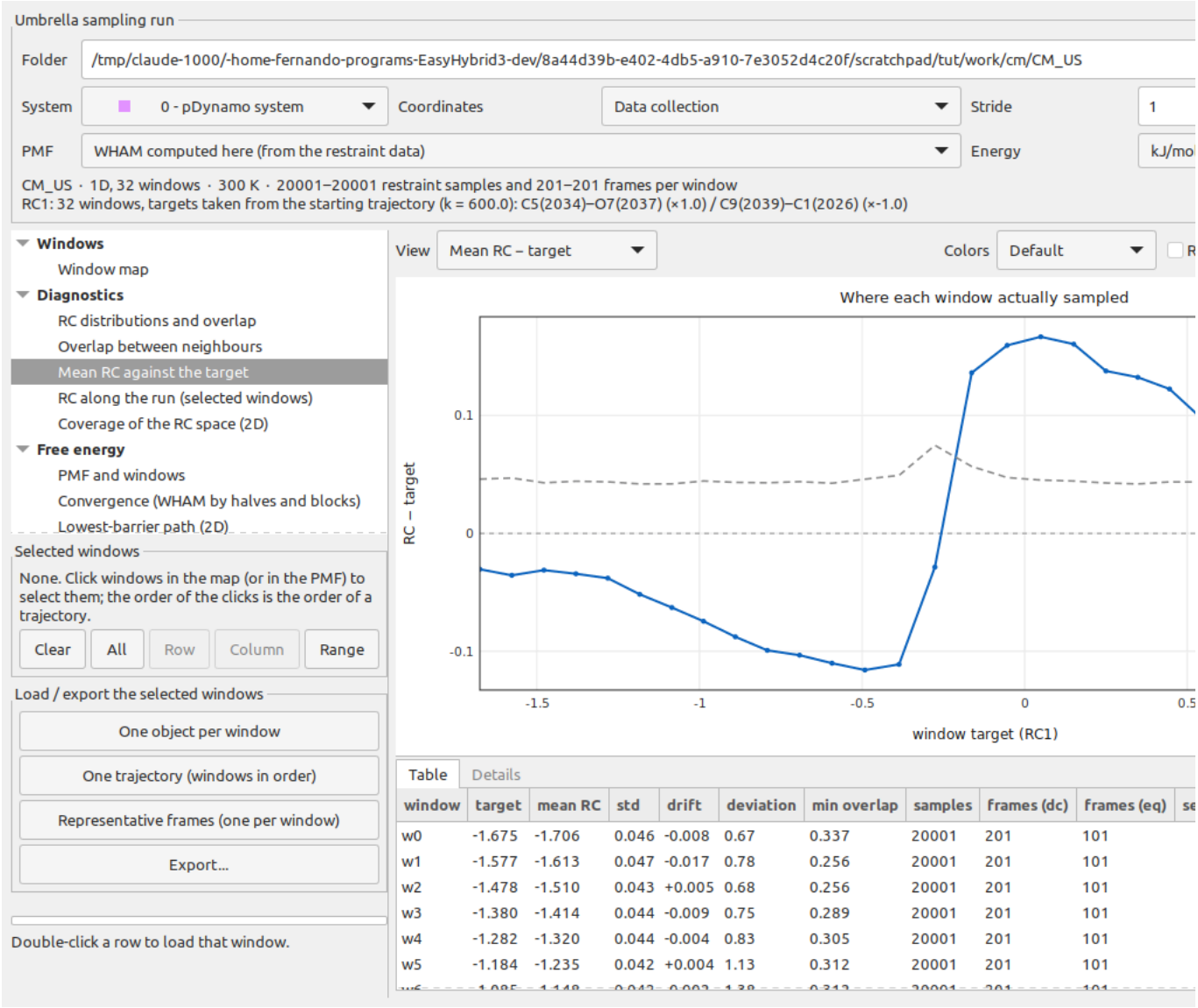



*Figure 3. Mean RC against the target: near the top the windows slide away from it.*

- The windows before the barrier sit below their targets and those after it above them, by up to 0.17 Å: on the steep sides of the barrier a restraint of 600 kJ mol$^{-1}$ Å$^{-2}$ cannot hold the coordinate, and each window slides downhill, away from the top.
- The Browser flags 4 windows (w16–w19) whose mean is more than 3σ from the target.
- The distributions leave the top (RC ≈ −0.3 to 0) covered by one broad window only.
- **Convergence (WHAM by halves)**: the PMFs of the first and second halves of the data differ by up to 12 kJ/mol.

The PMF of this run has a barrier of about 122 kJ/mol, but its top rests on very few samples.

## 2. Corrected protocol: a finer scan and k = 2000

1. Repeat the scan of part 1 with **Step Size 0.05 Å** and **Number of Steps 64** (2 minutes): 64 structures, the same profile (barrier 108.4 kJ/mol).
2. Umbrella sampling from it with **Force Constant 2000** kJ mol$^{-1}$ Å$^{-2}$ and 12 CPUs, the rest as before. σ = 0.025 Å, and windows 0.05 Å apart overlap by about 0.3; the restraint is now strong enough for the slopes of the barrier. The 64 windows took 2.4 h on 12 cores.

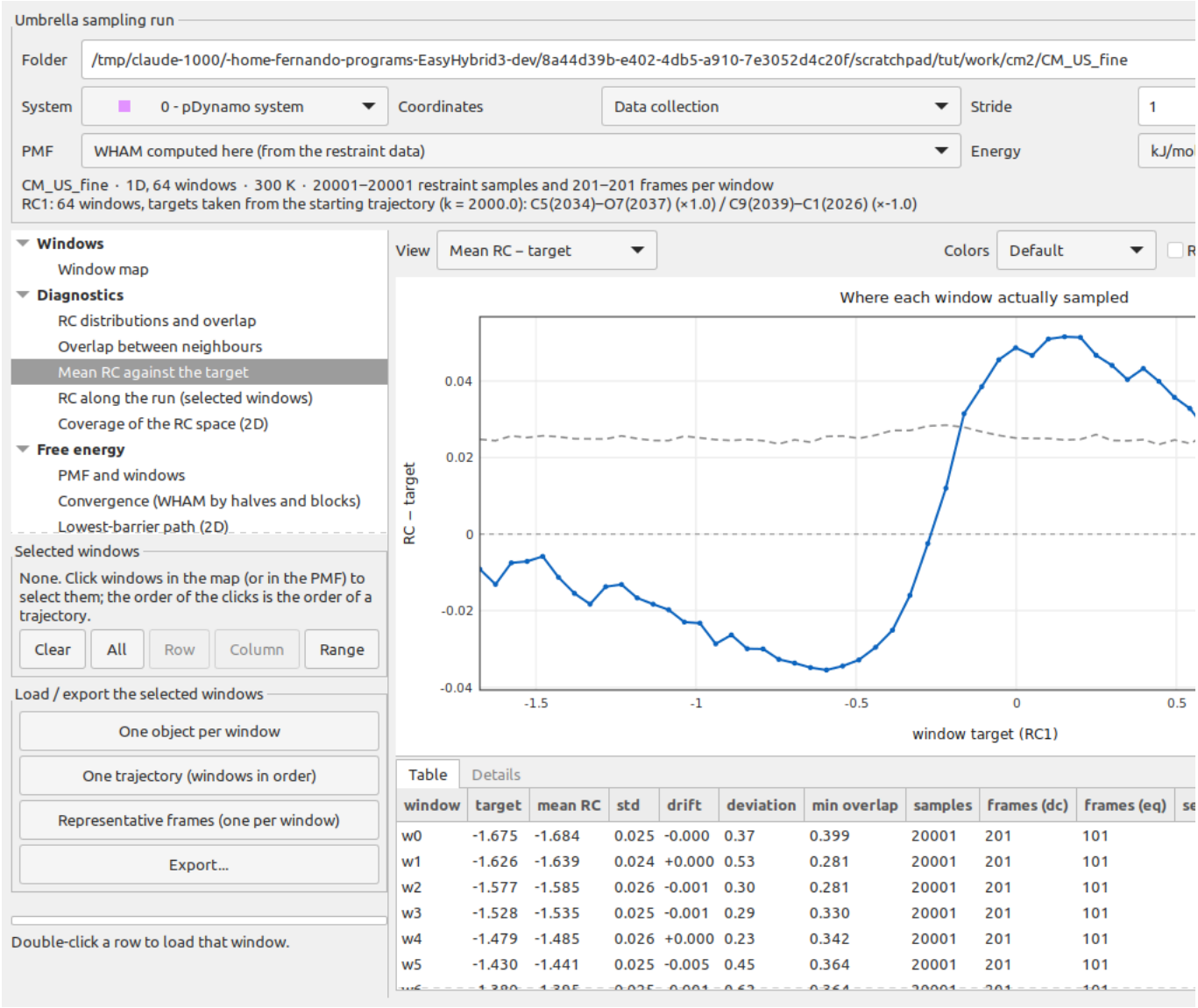



*Figure 4. Mean RC against the target, corrected protocol: all windows within 0.05 Å (2σ) of their targets.*

The Browser now reports no window more than 3σ from its target and no pair with an overlap below 0.03; the two halves of the data give PMFs within 7.4 kJ/mol.

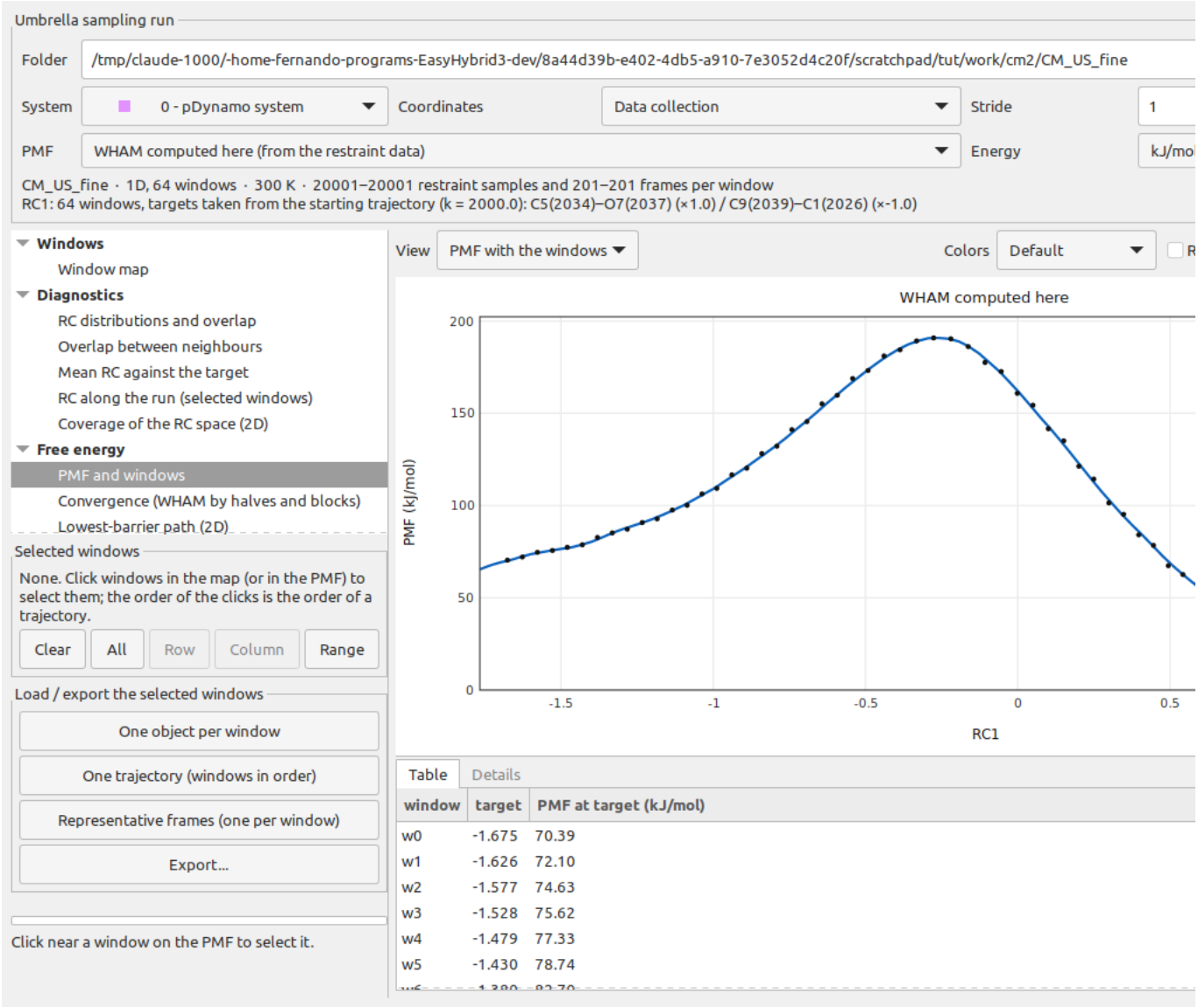



*Figure 5. The PMF of the Claisen rearrangement in CM (PM6/AMBER, 300 K).*

| | RC (Å) | PMF (kJ/mol) |
|---|---|---|
| Chorismate (first window) | −1.68 | 0 |
| Top | −0.26 | **120** |
| Prephenate (last window) | +1.55 | −70 |

## 3. What the numbers say

| | Barrier (kJ/mol) |
|---|---|
| Relaxed scan (part 1), potential energy | 108 |
| NEB (part 2), potential energy | 110 |
| Umbrella sampling, free energy | 120 |
| Experiment, *B. subtilis* CM ($k_{cat} \approx 50\ s^{-1}$) | about 64 |

The free energy barrier is about twice the experimental one. The main error is the method: PM6 overestimates the barrier of this rearrangement, and a QC region made of the substrate alone leaves the polarization of the active site to fixed MM charges. The usual remedy is to correct the PMF with higher-level energies (for example DFT single points on structures of the windows, or **Analysis → Energy Refinement** of the scan), or to use a better QC method (DFTB3, xTB, DFT) for the windows. The protocol lesson does not depend on the method: **check the windows before trusting the PMF**.

## Notes

- **Spacing and force constant go together**: for windows Δ apart on a flat surface, overlap ≈ erfc(Δ / 2√2σ) with σ = √(k_BT/2k). On a barrier the restraint must also beat the slope: 2k·(distance from the target) must exceed the force of the surface.
- **Cost.** QC/MM windows are expensive; run the data collection in parts (Export Script to a cluster), and look at *Convergence (WHAM by halves and blocks)* to decide whether more is needed.
- See also the tutorial QC/MM free energy of the SN2 step in LinB, with the same kind of diagnosis.

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EasyHybrid 3.2.0 · built 2026-10-29 from `dev` @ `3539da4` · https://github.com/ferbachega/EasyHybrid3